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Predictive models for inventory optimization: a machine learning application for demand forecasting at a construction supplies distributor

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Abstract

Inventory management is a critical task for distribution companies, as forecasting errors can lead to excess inventory, higher storage costs, or stockouts. This practice and policy paper presents a single-case study describing the development and deployment of a machine learning-based predictive model designed to optimize inventory management for a medium-sized Brazilian construction supplies distributor. Using the Amazon Forecast and Amazon SageMaker Canvas platforms, the study demonstrated how advanced demand forecasting techniques can potentially reduce stockouts and enhance operational efficiency. The research compared various predictive algorithms and assessed their performance using metrics such as RMSE, MAPE, MASE, and WAPE. The model achieved a weighted absolute percentage error (WAPE) of approximately 0.69% during the evaluation period, with performance varying across SKUs and demand patterns. The evaluation employed an 80/20 train-test split with a 60-day forecast horizon. Rather than claiming algorithmic novelty, this study's primary contribution is architectural and socio-technical since it documents an end-to-end implementation blueprint for deploying cloud-based AutoML in an SME that previously relied solely on descriptive analytics. The reliance on a single aggregate metric, the absence of baseline comparisons, limited transparency into proprietary AutoML, and the lack of quantitative measurement of business impact are discussed in the limitations of the study. The study details the technology architecture, data governance workflow, and human–AI interaction protocols, providing technical and managerial recommendations for SMEs seeking to transition from retrospective reporting to simplified predictive analysis, even with limited IT resources and without specialized Data Science teams.

Keywords Predictive models, Artificial intelligence, Machine learning, Inventory management, Supply chain management (SCM)

Context

Efficient inventory management is a vital challenge for distribution companies, as it directly affects customer satisfaction, operating costs, and overall business profitability. In the construction supplies industry, which features a diverse range of products and seasonal demand fluctuations, forecasting is particularly challenging [20]. Forecasting mistakes can result in excess inventory, raising storage costs and the risk of obsolescence, or lead

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to stockouts (product shortages), causing lost sales and customer dissatisfaction [8].

In this context, the objective of this practice paper is to develop and apply machine learning (ML) techniques to improve inventory management. The focus was on more accurate demand forecasting, cost reduction, and resource utilization. The practical application occurred in Casa Cardão, a medium-sized Brazilian distributor operating in both wholesale and retail channels for construction supplies. This business context highlights logistical and operational challenges, justifying the implementation of advanced predictive techniques to reduce disruptions and lower inventory holding costs.

As is typical in medium-sized organizations, the company has a lean IT (Information Technology) structure and lacks analysts with advanced expertise in Data Science and AI (Artificial Intelligence). Many available software solutions require significant investment and specialized staff. Currently, there is a shortage of these professionals in Brazil, so advanced applications of technologies such as AI and data analytics remain out of reach for many medium-sized Brazilian companies.

The inherent complexity of supply chain management (SCM) and demand forecasting, especially in scenarios with intermittent or irregular demand [21], justifies exploring more advanced methods than traditional ones, which are often limited in their ability to capture nonlinear patterns and multiple influencing variables [15]. The use of ML in SCM has emerged as a promising approach, providing powerful tools for analyzing large datasets and identifying complex patterns [22].

This practice paper addresses the following research question: How can cloud-based AutoML platforms enable SMEs (Small and Medium Enterprises) without specialized data science teams to implement effective demand forecasting systems for inventory management?

The study offers three contributions:

(i) Practical contribution: A detailed implementation blueprint documenting the end-to-end process of deploying AutoML-based demand forecasting in an SME, including data pipeline, model selection, and dashboard integration;

(ii) Methodological contribution: Documentation of how AWS AutoML workflows can be applied to time-series forecasting without requiring deep machine learning expertise;

(iii) Contextual contribution: Empirical evidence from a Brazilian SME operating in the construction supplies distribution industry, addressing a gap in the literature regarding AutoML adoption in emerging market SMEs.

The contribution of this study is primarily architectural, organizational, and pedagogical rather than methodological or algorithmic. The originality lies not in the algorithms used—which are standard and proprietary—but in the contextualized documentation of feasibility, governance, and implementation constraints when transitioning an SME from purely descriptive analytics (analyzing past data through reports) to a simplified predictive analytics capability. For organizations that have no forecasting tools beyond historical ERP (Enterprise Resource Planning) reports, as is the case studied in this paper, this leap may represent a significant and pragmatic capability gain, enabling autonomous future analyses without permanent dependence on external research teams.

Casa Cardão was selected as a representative case of a medium-sized Brazilian distributor facing inventory management challenges with limited IT resources. The company's characteristics—diverse product portfolio (10,500 + SKUs), seasonal demand patterns, lean IT structure, and regional market position—are typical of many SMEs in the construction supplies industry. The boundary conditions of this study include: (i) the Brazilian economic and regulatory context; (ii) the construction supplies industry with its specific demand patterns; (iii) availability of 16 + months of historical transactional data; and (iv) access to temporary AWS cloud credits. The implementation blueprint and governance workflow are expected to be generalizable to similar SME contexts, while specific accuracy metrics are context-dependent and should not be extrapolated without validation.

This practice paper outlines the process of developing, implementing, and evaluating ML-based predictive models using Amazon Web Services (AWS), specifically Amazon Forecast and Amazon SageMaker Canvas. This practical study demonstrates the feasibility and benefits of using these cloud-based technologies to improve inventory management decision-making, supporting the digital transformation and operational efficiency of an organization with a less structured IT department. The study also offers a reproducible model for Small and Medium-sized Enterprises (SMEs), enabling the use of machine learning for inventory management at lower costs.

The study was designed to develop an ML-based solution that integrates internal data from ERP (Enterprise Resource Planning) and CRM (Customer Relationship Management) business systems with external data (public data and partner databases) in a Data Lake environment. This practical application aims to provide more accurate forecasts, support informed decision-making, and reduce waste [16, 23].

The project was selected and received support through a specific innovation call for proposals from the National

Council for Scientific and Technological Development (CNPq), an entity associated with the Brazilian Ministry of Science, Technology, and Innovation (MCTI). The purpose of the call for proposals was to promote collaborations between organizations and universities in postgraduate research. The research project team included the project coordinator (a professor in Management), a PhD researcher in Management, and two undergraduate research scholarship students in Software Engineering.

Other partners were also involved in the project: i) a Brazilian business school, which provided access to its customer database to identify potential opportunities for testing and proof of concept; ii) a Paraguayan-Brazilian technological park associated with one of the largest hydroelectric power plants in the world, which supported submitting the research project to the FINEP (Brazilian Funding Agency for Studies and Projects, also associated with MCTI) call for innovation proposals, potentially enabling additional resources; iii) a cloud computing company based in São Paulo, which hosted the resources required for the project's development and infrastructure; iv) AWS—Amazon Web Services, which supplied a testing environment for over 12 months, enabling the execution of the POC—Proof of Concept; and v) a consulting and development firm in Minas Gerais, which provided access to and resources for AI development tools such as ChatGPT Pro, Manus AI, and Base44.

The project's overall goal was to research, identify, and develop ML-based models and technological solutions that can work as predictors and automate data collection and analysis using Artificial Intelligence. The project aimed to create an intuitive interface for managers that enables accurate simulations and forecasts for inventory management. A specific goal was to make application development easy to duplicate at low cost, using open-source software or no-code or low-code platforms (with little or no coding or development), enabling its adoption by companies with reduced IT budgets and personnel.

Beyond this introduction, the next section of this report presents the theoretical background that support the scientific framework that guides the study's development. Sect. "Application development" details the application development, reports the experiment conducted, presents the achieved results, and discusses the application of the tools. Sect. "Analysis of the results" evaluates the study across the following dimensions: i) how well it aligns with related fields and areas of Management; ii) the potential environmental and stakeholder impacts of the project; iii) the feasibility of applying the proposed technology and processes; iv) the innovation and knowledge level involved in product development; and v) the complexity of the network and relationships necessary for the project design and execution. Sect. "Discussion of results" discusses the results, while Sect. "Conclusion" concludes the report with recommendations for future work and addresses its limitations.

Theoretical background

This section provides an overview of the theoretical framework that guided the application development. Inventory management is central to logistics and operations management, balancing inventory costs with the risk of product shortages. Accurate demand forecasting is essential for effective inventory control [9]. Traditional forecasting techniques, such as moving averages, exponential smoothing, and ARIMA (Autoregressive integrated moving average) models, are widely used but may have limitations when handling complex, seasonal, or multivariate data [5].

Demand in the retail and distribution industry, such as construction supplies, often exhibits characteristics like intermittency (periods without demand) and seasonality. Traditional models may struggle to capture these patterns accurately [20, 21]. Additionally, external factors such as promotions, economic conditions, changes in tax laws, political events, competitor activities, and even weather can influence demand, requiring models that can incorporate multiple sources of information.

Predictive models using Artificial Intelligence (AI) for inventory management offer significant advantages over traditional methods. ML algorithms process many factors and identify complex patterns that traditional techniques overlook [3], resulting in more accurate demand forecasts and enabling quick adaptation to changing market conditions [12]. As a subfield of AI, ML provides alternative and complementary methods to traditional statistical forecasting techniques. ML algorithms can identify complex patterns in historical data, including nonlinear relationships and interactions among multiple variables, without relying on strict statistical assumptions [15].

Product demand forecasting is an essential technique for inventory management, enabling companies to optimize stock levels, reduce costs, and enhance customer service. One machine learning method that has gained prominence in this field is random forest for forecasting [24]. Other machine learning algorithms have been used effectively for demand forecasting, including artificial neural networks (ANNs), support vector machines (SVMs), and decision tree-based methods such as random forest and gradient boosting [22, 24]. These approaches are especially valuable when large amounts of

data (Big Data) are available and demand is driven by a complex mix of factors [6].

Multiple studies show that this technique (Random Forest) is highly effective for inventory management. Van Steenbergen and Mes [23] introduced DemandForest, an innovative approach that combines K-means clustering, random forests, and quantile regression forests to forecast demand patterns for new products. The method was tested on real datasets and outperformed several reference methods, resulting in about 15% savings in inventory [23]. Vairagade et al. [24] compared various demand forecasting models, including random forest and artificial neural networks. They found that random forest was more accurate, making it better suited for predicting product demand in complex supply chains.

On the other hand, Tian et al. [21] addressed intermittent demand forecasting, a common challenge in retail, and proposed a combined approach that incorporates current inventory and past sales. This method, known as the Markov-combined method (MCM), proved more accurate than traditional techniques, leading to significant improvements in inventory management.

Punia et al. [17] introduced another combined approach that uses long short-term memory (LSTM) and random forest models to capture complex temporal and regression relationships. This method significantly outperformed other forecasting techniques in accuracy and robustness in multichannel retail settings.

Naik et al. [16] examined sales forecasting in the food industry using random forest regression. This model was notable for its high accuracy compared with other machine learning methods, making it well suited for predicting sales and optimizing inventory management.

In summary, the reviewed literature highlights three key pillars that guided the application design: (i) the superiority of ML approaches over traditional statistical methods in volatile demand scenarios, due to their ability to capture nonlinear relationships and multiple variables [9, 15],(ii) the need to combine algorithmic forecasting with human judgment to enhance operational decision-making, as supported by studies on managerial adjustments in forecasting (Goodwin & Fildes, 2019); and (iii) the importance of explainability mechanisms that clarify the most influential factors behind the forecast, thereby increasing managers' confidence in the system's recommendations [19].

It is important to emphasize that the majority of the cited studies evaluate algorithmic performance (i.e., forecasting accuracy of specific ML methods), whereas relatively few studies examine the implementation process, organizational adoption, or the integration of ML-based forecasting into decision making in practice. This distinction is critical for positioning the present study, whose contribution lies primarily in the domain of documenting the feasibility of implementation, governance, and socio-technical integration, rather than in advancing algorithmic performance per se.

The deployment of automated forecasting systems raises important data governance questions. Organizations must weigh the benefits of automation against the risks of over-reliance on algorithmic recommendations [19]. Key governance considerations include: (i) Transparency and explainability: Understanding how forecasts are generated and which factors drive predictions; (ii) Monitoring and accountability: Establishing responsibility for forecast quality and decision outcomes; (iii) Data quality and bias: Ensuring input data integrity and detecting potential biases in historical patterns; (iv) Human oversight: Defining appropriate levels of human intervention and override capabilities; and (v) Vendor lock-in: Managing dependencies on proprietary platforms and ensuring auditability. These data governance aspects are particularly critical for SMEs, which may lack the technical capacity to audit complex ML systems independently. While ML forecasting brings potential benefits, a significant adoption gap exists for SMEs since traditional ML implementation requires: (i) data science expertise for algorithm selection, feature engineering, and hyperparameter tuning; (ii) software engineering capabilities for production deployment; (iii) computational infrastructure; and (iv) ongoing model monitoring and maintenance [7].

AutoML platforms aim to democratize ML by automating algorithm selection, hyperparameter optimization, and model evaluation [11]. Cloud-based AutoML services (e.g., AWS SageMaker, Google AutoML, and Azure ML) further reduce infrastructure barriers by providing managed environments with prebuilt pipelines.

However, research on AutoML adoption in SMEs remains limited since most studies focus on algorithmic performance rather than on organizational feasibility, implementation challenges, or data governance requirements. The socio-technical aspects of AutoML adoption—including user trust, interpretability, and integration with existing decision processes—are particularly underexplored [19].

These findings informed the selection of AutoML platforms that integrate diverse algorithms (e.g., DeepAR + and random forest) and robust evaluation metrics. Although the literature recognizes the potential benefits of ML for demand forecasting in complex environments, several gaps remain. First, most empirical studies focus on large enterprises with dedicated data science teams, leaving limited evidence on AutoML

adoption in SMEs (particularly in emerging markets). Second, there is uncertainty about the practical implementation challenges, governance requirements, and total cost of ownership for cloud-based ML solutions in resource-constrained organizations. Third, the human–AI interaction protocols needed for successful adoption in organizations lacking technical expertise remain underexplored. This practice paper addresses these gaps by documenting an end-to-end implementation of cloud-based AutoML for demand forecasting in a Brazilian SME, including the technology architecture, data governance workflow, and lessons learned.

As detailed in the next section, these theoretical guidelines were put into practice in the application development, defining the data pipeline, model selection criteria, and human–machine interaction protocol that form the foundation of the implemented solution.

Application development

This section describes the application development process, covering the IT tools, the underlying architecture, and the performance assessment. The application context is Casa Cardão, a hundred-year-old Brazilian distributor operating in both wholesale and retail channels for construction supplies. It employs around 220 staff members and 200 sales representatives. It operates two regional distribution centers covering 12,000 m^2 of storage and markets a portfolio of over 10,500 SKUs (Stock Keeping Units), with strong demand for 150 items from the ABC curve. In 2023, the company generated R$310 million (about US$60 million) in revenue and held a 4% market share in Minas Gerais state within the cement, mortar, and ceramic tile segments [18].

Development tools

Developing and deploying ML models can be complex and require substantial computational resources. Cloud platforms such as Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure offer managed services that can simplify this process. In this study, two AWS tools were used. Amazon Forecast is a fully managed service that uses ML to deliver highly accurate time-series forecasts. It automates much of the ML process, from data loading to model training and deployment, automatically testing various algorithms (including ARIMA, ETS, Prophet, DeepAR+, and others) and selecting the best one for the data provided [1]. In July 2024, AWS closed new customer access to Amazon Forecast, directing users to SageMaker Canvas as the recommended alternative. Existing Amazon Forecast customers can continue using the service, but AWS recommends transitioning to SageMaker Canvas for enhanced features and cost efficiency. The second tool, Amazon SageMaker Canvas, is a no-code visual interface that enables business analysts to build ML models without writing code. It streamlines data preparation, model training, and forecast generation, making ML more accessible to non-expert users [2]. These platforms facilitated efficient experimentation and implementation of predictive models in the Casa Cardão environment.

The project produced a dashboard or visual layer that displays structured data collected through an ETL (Extract, Transform, and Load) process using AWS SageMaker Data Wrangler, covering both external and internal data. The dashboard includes a series of graphs showing inventory forecasts (Fig. 1). A prompt feature on the generative AI-powered dashboard interface provides detailed explanations of the methodology and graphs used, making the tool more user-friendly for managers.

Additionally, the interface enables managers to run simulations as needed, combine data from multiple sources, and leverage this information to enhance inventory management. The project is delivered in a tailored manner, using company-specific data along with information from public sources. The conversational management dashboard can be continuously improved by adding new data sources to develop a more robust and effective predictive model for inventory management.

The developed product is a demand prediction system that uses predictor results generated by AWS SageMaker Canvas, featuring an interactive dashboard that provides product demand forecasts across three scenarios: optimistic, neutral, and pessimistic. In the optimistic scenario, inventory levels are low, and the forecast accounts for seasonality and promotions. In the pessimistic scenario, inventories are high, and there is no forecasted demand for the products. The neutral scenario indicates no discernible trend in demand, with stock levels remaining balanced. The system's features include time-series analysis, integration of internal and external data, customized simulations, and a console that explains the methodology and the graphs used.

The project can help the company by lowering costs related to excess inventory or product shortages and by making inventory management more efficient, enabling more confident decision-making. Specifically, the goal was to create AI-based inventory management software that could predict necessary inventory levels with 90% accuracy, and the model achieved a WAPE of approximately 0.69% during the evaluation period.

However, this aggregate metric should be interpreted with caution, as WAPE is volume-weighted and may be disproportionately influenced by high-turnover SKUs, potentially masking poorer performance on low-volume

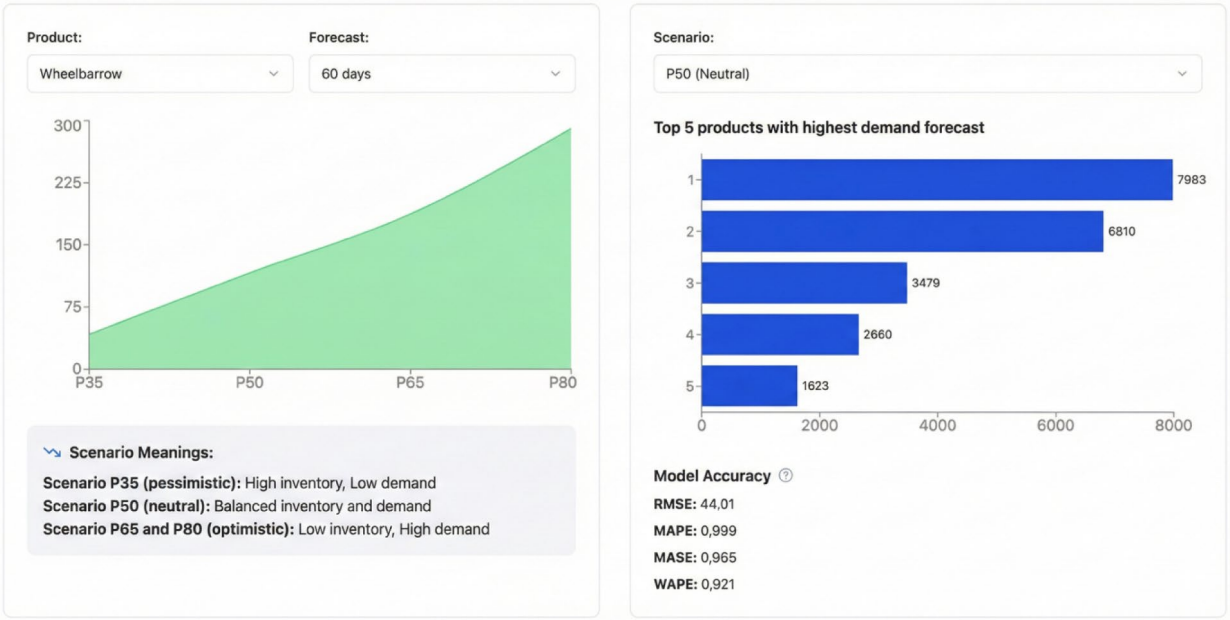


Fig. 1 Inventory forecast graphs. *Note*: Screenshots of the dashboards provided by the demand forecast application

or intermittent demand items. Operational objectives include reducing excess inventory by 20% and shortages by 15% within 1 year of implementation. Preliminary qualitative observations suggest positive trends, though systematic measurement of business impact—including quantitative assessment of inventory cost reduction, service-level improvements, and return on investment—will require a longer post-deployment evaluation period and was not conducted within the scope of this study.

Several machine learning techniques were tested and used to develop the project, including AWS's proprietary algorithm, DeepAR. It is based on recurrent neural networks (RNNs), with an LSTM (Long Short-Term Memory) model, and is often used to generate probabilistic forecasts for multiple series (across various products). DeepAR demonstrated the best accuracy among the models available on the AWS platform, including DeepAR+, CNN-QR, Prophet, ARIMA, and ETS.

Technology architecture and phases

Application development began in August 2024, and by April 2025, the proof of concept (POC) and validation of the results were complete. The initial POC was developed using Amazon Forecast, but due to the AWS's transition announcement, the production deployment migrated to SageMaker Canvas. The system was fully operational within the organization by July 2025. Figure 2 illustrates the application's technology pipeline, organized into four functional layers in the AWS environment. In the ingestion layer, transactional data extracted from Casa Cardão's ERP and CRM systems, along with external data such as weather, holidays, and macroeconomic indices, are stored in a data lake on Amazon S3. The preparation layer uses SageMaker Data Wrangler for ETL, ensuring the cleaning, aggregation, and enrichment of time-series data. Next, the modeling layer employs Amazon SageMaker Canvas for exploratory AutoML and the Forecast service (DeepAR+, CNN-QR, Prophet, among others) for final training, automatically selecting the algorithm with the lowest backtest error [1, 2]. The generated forecasts are stored in a versioned S3 bucket, and AWS Lambda functions manage daily model updates, providing serverless scalability.

At the application layer, the results are integrated into an interactive dashboard developed in React on the base44 platform. This dashboard displays P35 (pessimistic), P50 (neutral), P60, and P80 (optimistic) scenarios, complemented by a conversational interface powered by GPT O3-mini. This modular and fully managed workflow facilitates rapid iteration, cost efficiency, and seamless integration between algorithmic output and human decision making.

A set of technological, functional, and legal requirements was considered for the project development. Technologically, an adequate IT infrastructure was required to support the collection, storage, and processing of large volumes of data. AWS was selected for its resources, widespread availability, and ease of implementation. Functionally, the

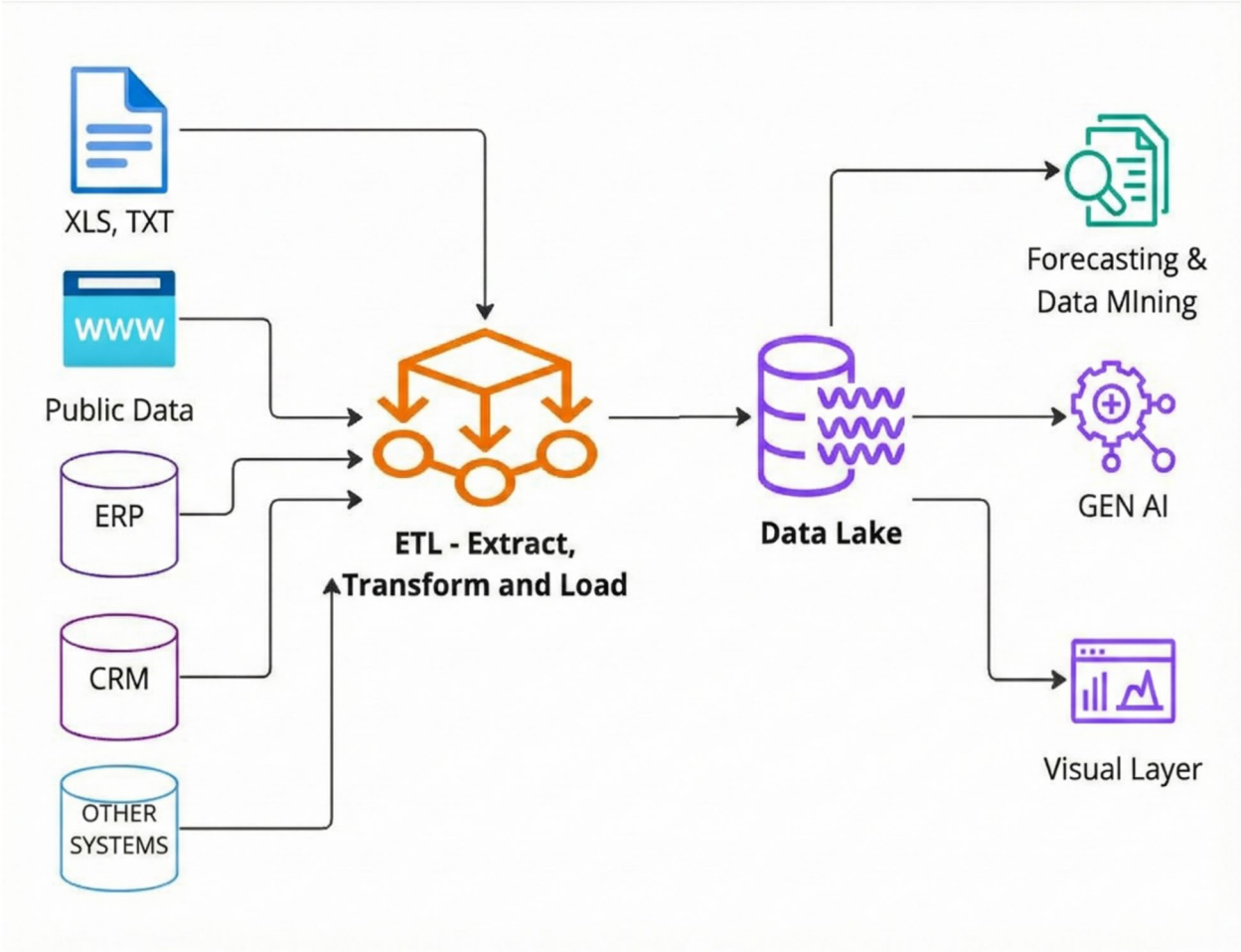


Fig. 2 Technological Architecture for the Development of the Demand Forecast *Application*

developed interface—dashboard—is intuitive and user-friendly, integrates with results generated by AWS, and provides scenario analysis features for users. Legally, the project complies with data protection and privacy regulations and does not handle any sensitive data.

The stakeholders included both internal and external parties. Internally, the research team and IT team at Casa Cardão, along with the project manager at the cloud computing company, were essential. Externally, AWS served as a key technology provider. AWS provided the credits and resources needed for testing through its cloud computing and AI infrastructure. The Proof of Concept (POC) was conducted at Casa Cardão, providing a real environment to test and assess the technology's effectiveness. The application development was structured into five distinct phases: planning, development, testing and validation, implementation, and evaluation and improvement. The planning phase included defining requirements and scope, as well as a detailed project schedule to ensure clarity regarding the overall objectives.

In the development phase, the predictor and an interactive dashboard were built. During the testing and validation phase, software testing, predictive model validation, and POCs were conducted. These first three phases occurred between August 2024 and April 2025.

Subsequent phases include implementation, which comprises implementation itself, user training, and initial support; and finally, evaluation and improvement, which involves performance monitoring, collecting user feedback, and continuous improvement. These last two phases were completed by July 2025. Figure 3 details each of these phases.

The main challenge was to improve the accuracy of demand forecasting at Casa Cardão to enhance inventory management. The initial phase involved gathering historical sales data from the company, covering a relevant period (usually 3 to 5 years, depending on

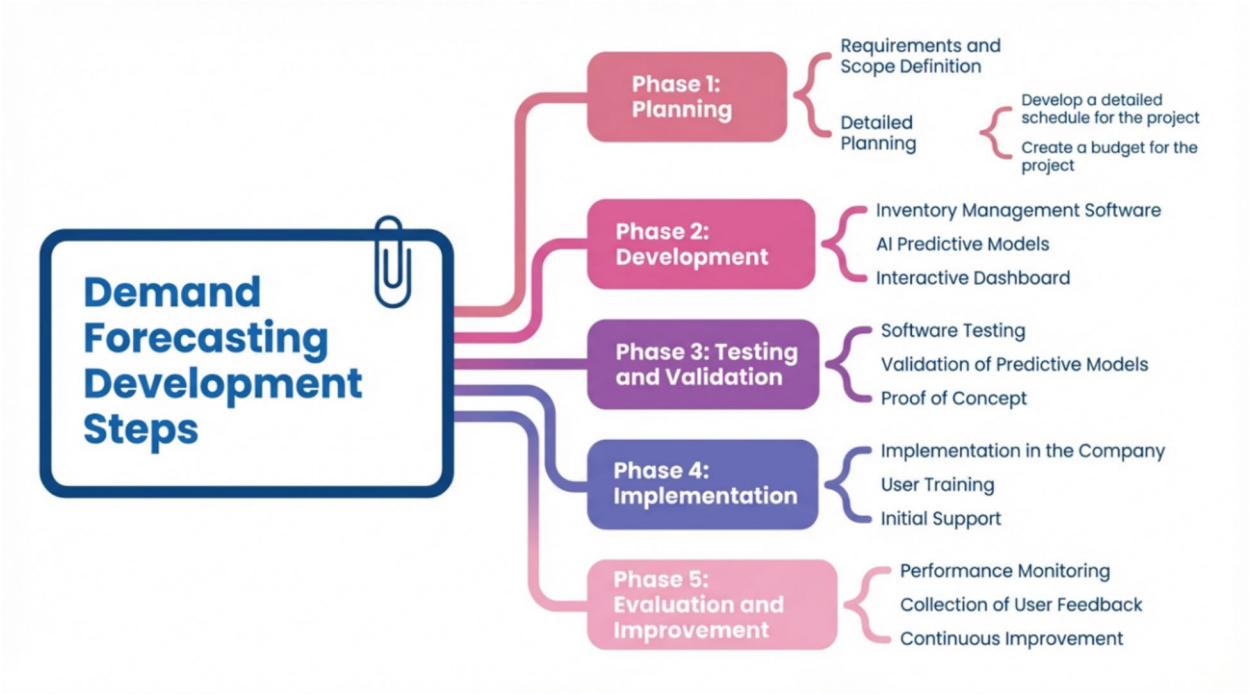


Fig. 3 Phases for development of the demand forecasting application

availability and detail) to identify seasonal patterns and trends. The data included, at a minimum, product identifiers (SKUs), transaction dates, and quantities sold. Additional information, such as details on promotions, holidays, or product characteristics (metadata), was also considered to improve the models.

The collected data were cleaned and preprocessed. This procedure included handling missing values, aggregating the data to the required forecast level (e.g., daily or weekly sales per SKU), and identifying and addressing outliers as needed. The target time series (sales per SKU per period) and related datasets (item metadata and related time series, such as promotion data) were created.

The Amazon Forecast and Amazon SageMaker Canvas platforms were used to develop and train the predictive models. The prepared data were uploaded to the service. Amazon Forecast automatically trained and evaluated multiple time-series forecasting algorithms, including DeepAR+, CNN-QR, Prophet, ARIMA, and ETS. The service selected the best-performing algorithm based on predefined evaluation metrics (backtesting). Predictors (trained models) were generated for the SKU set. However, in the second half of 2024, AWS discontinued the platform. A complementary or alternative approach was explored using SageMaker Canvas as a natural replacement for Amazon Forecast. The data were imported into the visual interface, and predictive models focused on time-series forecasting were built using the platform's automated features, enabling users with less coding expertise to quickly conduct feasibility assessments.

Methods and validation protocol

Table 1 summarizes the dataset characteristics and evaluation protocol, detailed as follows.

Data Characteristics. For the exemplified SKU ("Amanco 6 m×100 mm drain pipe"), the historical data comprised 16 months of daily records (July 2023 – November 2024), totaling 11,074 observations. Data were aggregated to daily granularity. The dataset was split into training and backtesting sets at an 80/20 ratio. Missing

Table 1 Summary of Dataset Characteristics and Evaluation Protocol

Characteristic	Description
Time span	July 2023—November 2024 (16 months)
Observations	11,074 daily records (exemplified SKU)
Granularity	Daily
Train-test split	80% training/20% backtesting
Forecast horizon	60 days
Algorithms compared	DeepAR+, CNN-QR, Prophet, ARIMA, ETS
Selected model	DeepAR+ (lowest backtest error)
Primary metric	WAPE (Weighted Absolute Percentage Error)

values were handled using SageMaker Data Wrangler's automated imputation procedures. Outliers were identified using statistical thresholds but retained in the training data to preserve demand variability.

Features: Internal variables included historical sales quantities from the ERP system and customer information from the CRM. External variables included weather data, holiday calendars, and macroeconomic indices. Feature inclusion was informed by domain expertise and the literature; however, a formal ablation study comparing model performance with and without external features was not conducted. A feature importance analysis to isolate the contribution of individual variables represents an avenue for future research.

Validation Protocol: AWS AutoML performed backtesting by splitting the time series into training and testing sets. The platform automatically trained and evaluated multiple algorithms (DeepAR+, CNN-QR, Prophet, ARIMA, and ETS) and selected the model with the lowest backtest error. Hyperparameter tuning was handled automatically by the AWS AutoML system; specific hyperparameter values were not manually configured. The forecast horizon was set to 60 days to align with operational purchasing cycle requirements.

Model Selection: DeepAR+, a recurrent neural network algorithm developed by Amazon, was selected by the AutoML process as the best-performing model for the evaluated SKUs. The selection criterion was the lowest WAPE on the backtest dataset. We acknowledge that reliance on AWS AutoML automation offers both practical benefits for SMEs lacking data science expertise and methodological limitations regarding full transparency and reproducibility.

Evaluation was conducted on a set of 10 representative SKUs selected from the company's ABC classification. A systematic comparison with simple baseline models (naïve and seasonal naïve forecasts) was not performed because the AWS AutoML platform operates as a "black box" that does not expose its internal benchmarking process or allow users to inject custom baseline algorithms into its evaluation pipeline. The platform automatically tests its own suite of algorithms (including DeepAR+, CNN-QR, Prophet, ARIMA, and ETS) and selects the best performer; however, it does not report how these compare against elementary baselines such as naïve persistence. Similarly, advanced hyperparameter optimization (e.g., via grid search or genetic algorithms) was not manually performed, as the AutoML platform handles this process automatically. Indeed, this intrinsic design limitation is the core value proposition of AutoML services, representing a deliberate trade-off between methodological control and implementation practicality for resource-constrained SMEs. Error distribution by SKU class (e.g., ABC classification or intermittent vs. smooth demand) and by forecast horizon was not analyzed, representing a limitation of this study and an opportunity for future research.

Performance evaluation

The model's performance was assessed using standard time-series forecasting metrics, with backtesting conducted on AWS. The following metrics were considered:

RMSE (Root Mean Squared Error), which gauges the average size of errors and penalizes larger deviations [10]:

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \widehat{y}_t)^2}$$

where y_t is the actual demand at time t and $\widehat{y}_t$ is the predicted demand.

MAPE (Mean Absolute Percentage Error), which expresses errors in percentage terms relative to actual demand and is sensitive to low values [10, 14]:

$$MAPE = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{y_t - \widehat{y}_t}{y_t}\right|$$

MASE (Mean Absolute Scaled Error), which compares the model's performance against a naive model and is especially useful for intermittent data [10]:

$$MASE = \frac{\frac{1}{n}\sum_{t=1}^{n}|y_t - \widehat{y}_t|}{\frac{1}{n-1}\sum_{t=2}^{n}|y_t - y_{t-1}|}$$

MASE compares forecast error to a naïve baseline (previous period's value). MASE < 1 indicates better performance than naïve forecasting [10].

Weighted average percentage error (WAPE)

$$WAPE = \frac{\sum_{t=1}^{n}|y_t - \widehat{y}_t|}{\sum_{t=1}^{n} y_t} \times 100\%$$

where y_t is the actual value and $\widehat{y}_t$ is the forecast. WAPE weights errors by total demand, providing robustness for items with varying sales volumes [13]. The WAPE metric was chosen as the primary indicator because it reflects the impact on total sales volume and is less sensitive to outliers than RMSE. WAPE was selected as the primary metric because it is less sensitive to low-volume periods than MAPE [10] and provides an aggregate measure of forecast accuracy that weights forecast accuracy by demand volume [13].

After selecting the best-performing model(s), demand forecasts were generated for a relevant future period (e.g., the next 4–12 weeks). These forecasts were then provided to Casa Cardão's Commercial Department via reports and the dashboard to support inventory replenishment and purchasing planning decisions, and to validate the results.

The implementation of predictive models using Amazon Forecast and SageMaker Canvas at Casa Cardão produced important results. A comparative analysis of algorithms trained on AWS platforms showed that ML-based models, such as DeepAR + (a recurrent neural network from Amazon), often outperformed traditional statistical models (e.g., ARIMA and ETS) for most SKUs analyzed, particularly those with more complex or seasonal demand patterns.

To demonstrate the model's performance on a high-turnover item, the SKU "Amanco 6 m × 100 mm drain pipe" was selected, classified as "A" for value and "B" for volume in the Casa Cardão portfolio. The historical data comprised 16 months of daily records (July 2023 – November 2024), totaling 11,074 observations, with 80% used for training and 20% for backtesting. The forecast horizon was set to 60 days. In the optimistic scenario (P80), chosen because the factory delayed passing on the exchange rate increase and kept prices stable despite the US dollar's appreciation, the DeepAR + algorithm estimated 2,340 units for December 2023, while actual sales were 2,255 units, resulting in a WAPE of 0.7%. The deviation of less than 1% illustrates the model's performance for this specific high-volume item and highlights the synergy between statistical forecasting and managerial judgment in selecting the appropriate quantile for market conditions.

Developed through optimization and selection by Amazon Forecast, the final predictive model achieved a WAPE of approximately 0.69% during the evaluation period. This metric represents the aggregate weighted error across the evaluated SKUs. WAPE is volume-weighted, meaning that high-volume items contribute proportionally more to the aggregate measure [13]. Per-SKU class breakdowns (e.g., by ABC classification) and a separate analysis of intermittent demand items were not conducted in this study; these represent avenues for future analysis.

The flowchart (Fig. 4) illustrates the entire cycle of the predictive demand model, outlining seven macro-stages integrated within a cloud computing environment. The process begins with the automated extraction of transactional data from the company's ERP and CRM systems, along with external variables—macroeconomic indicators, holiday calendars, and weather data—which are stored in a data lake on Amazon S3. The seventh step involves continuous accuracy monitoring: a daily verification process compares actual sales with predictions; if the WAPE exceeds the 5% threshold, a new retraining cycle is automatically initiated, ensuring adaptive learning and predictive robustness over time. We acknowledge that the cold-start problem for new products (lacking sufficient historical data) has not yet been addressed in this implementation and represents an opportunity for future development.

Next, using SageMaker Data Wrangler, the ETL pipeline runs, responsible for cleaning, aggregating over

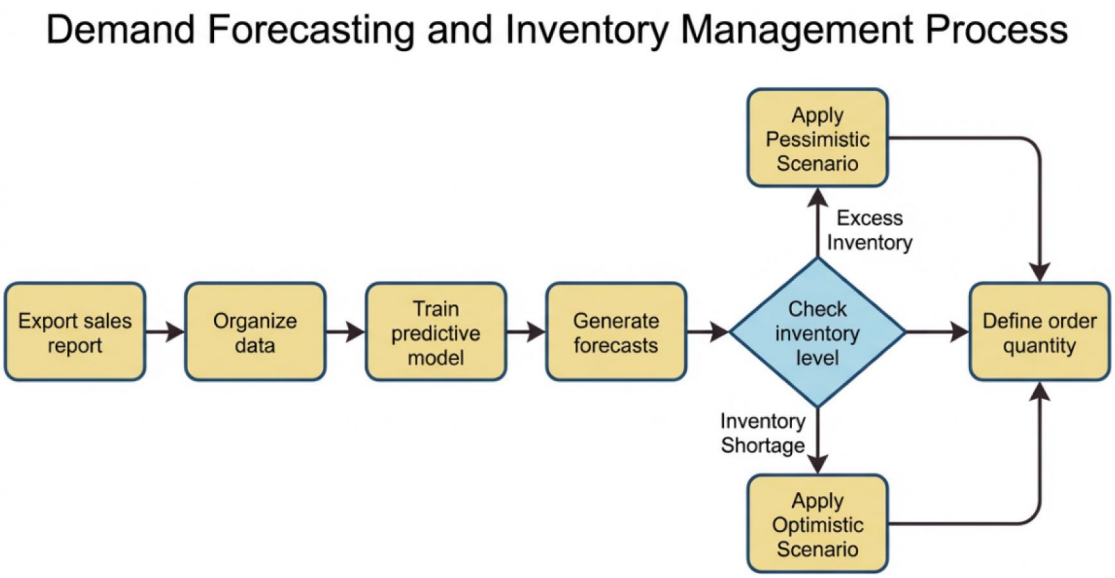


Fig. 4 Demand Predictor Model Flowchart. *Note*: Flowchart to be used during the inventory replenishment process

time, and enriching historical data to ensure its quality for modeling. In the third step, this structured data is fed into SageMaker Canvas, where multiple time-series algorithms are automatically trained and validated via backtesting; the model with the lowest weighted absolute error (WAPE) is then versioned as the active predictor.

The next step involves periodically generating forecasts, storing them in a separate S3 bucket, and simultaneously sending them to a relational database via AWS Lambda functions for real-time access. In the fifth step, these forecasts are used by an interactive dashboard built with React (an open-source JavaScript library for building user interfaces), where the manager selects P50, P60, or P80 scenarios and reviews the key factors influencing each forecast horizon. The resulting decisions—such as adjustments to orders, promotions, or inventory policies—are fed back into the system in the sixth step by recording the actions taken and their outcomes, thereby forming an operational feedback database.

The seventh step involves continuous accuracy monitoring: a daily verification process compares actual sales with predictions. If the WAPE exceeds the 5% threshold, a new retraining cycle is automatically initiated, ensuring adaptive learning and predictive robustness over time. This modular and coordinated chain offers scalability, data governance, and alignment between algorithmic intelligence and managerial judgment, which are essential for the successful adoption of demand models in small and medium-sized enterprises.

During the POC, the primary results from implementing predictive models showed significant potential benefits for the business, particularly a reduction in stockouts, enabled by more accurate forecasts that improved replenishment planning. Additionally, inventory levels were optimized by adjusting safety stocks and reorder points, leading to better allocation of working capital by balancing availability and excess inventory. Purchasing planning also improved, with forecasts providing strategic support for supplier negotiations and greater alignment among procurement, inventory, and demand.

The adoption of AWS platforms streamlined the development and implementation of models, enabling the team to focus on analyzing results and integrating forecasts into business processes, despite initial limitations in deep machine learning.

Analysis of the results

This section provides detailed information on key aspects of the application, including its compliance with management areas, its impact on organizational and social aspects, and its usefulness across different business environments. Additionally, it examines the advancements brought by integrating predictive models with ML tools, emphasizing how these innovative solutions can change decision making and improve processes. Lastly, it discusses the complexity and importance of these models for small and medium-sized businesses.

Adherence

The application of predictive models and inventory management is closely connected to various areas of management and related disciplines, with a special focus on the management of information systems. Predictive models align with Theme 1 on decision making (theories, modeling, support systems, and technologies), promoting tools that support and facilitate decision making in complex environments. Additionally, the application relates to digital transformation and innovation, enabling greater efficiency in inventory and operations management. The application also engages with Big Data, Data Science, and AI for Strategic Intelligence, as ML applications require large volumes of data. Beyond technology and information, the application links to other management fields, including operations and logistics, innovation, marketing, and finance.

Impact

Developing a ML project to predict product demand can significantly improve how companies manage complex inventories. This influence extends across the organization, community, and local area, helping to improve operational efficiency, reduce costs, and boost customer satisfaction.

The application aimed to promote AI adoption among small and medium-sized enterprises (SMEs) through a range of strategies. First, the application focused on providing user-friendly, pre-made AI tools tailored to SMEs' specific needs, which can be easily integrated into existing systems and workflows. This feature reduces the need for specialized AI technical expertise. Additionally, the solution automates routine and repetitive tasks, freeing companies to allocate time and resources to activities of greater strategic importance, thereby enhancing operational efficiency and reducing costs.

The present application can benefit the environment by preventing the loss of perishable items and reducing unnecessary consumption in the food supply chain, thereby making resource use more efficient and lowering carbon emissions. For example, it can help companies better manage their inventories and supply chains, preventing material waste through demand forecasting, delivery route optimization, and real-time inventory monitoring. The application can also have a significant

positive social impact by promoting greater efficiency and growth for companies. The automation of routine tasks and the improved decision making enabled by AI can boost business productivity.

Applicability

The replicability of this type of project in other contexts is high owing to the adaptability of ML algorithms. Small and medium-sized (SMEs) businesses can also use cloud-based ML services, such as AWS, Google AI, and Azure, which provide scalable and cost-effective solutions. Especially for SMEs, considerable potential for applying the proposed model is evident, as low-code platforms do not require highly skilled machine learning teams.

ML algorithms can be integrated with ERP systems and other business management platforms, enabling companies to use historical and real-time data to train models and improve forecast accuracy. The scalability of these models is a key benefit, as they can adapt to the volume and complexity of inventory operations. ML's ability to detect intricate patterns and generate precise predictions greatly reduces human error and streamlines inventory and replenishment processes, especially in business environments where demand is volatile and unpredictable.

ML algorithms are sufficiently adaptable to operate across diverse settings and can be tailored to manage inventory across industries such as retail, manufacturing, and distribution. Thorough documentation of models, training procedures, and integrations supports replication in new environments without requiring a restart.

The proposed application is a modular solution that enables specific components to be reused or adapted for various applications, such as modifying a demand forecasting model to predict sales across different markets or products. The potential market for the project is extensive, as any company with critical inventory management needs can benefit from predictive models. In the industrial sector, these models can forecast demand for specific products, optimize production, and reduce excess inventory. Additionally, they can identify products at high risk of becoming obsolete, enabling preventive measures, and predict equipment failures to facilitate maintenance and avoid production downtime. Distributors and wholesalers can use these models to optimize inventory levels, prevent shortages, and minimize storage costs. They can also forecast demand for seasonal products to ensure adequate stock and identify slow-moving items to inform discontinuation decisions. In retail, predictive models help forecast demand for particular products in specific stores, preventing shortages and maximizing sales. They also identify high-turnover products to optimize store displays and predict the end of product life cycles, aiding promotions and reducing waste.

For service companies, predictive models help forecast demand for specific services, optimize resource allocation, and prevent overload. They can identify customers at high risk of default, enabling preventive measures and predict equipment maintenance needs to avoid service interruptions.

Innovation

The innovation contribution of this application lies not in algorithmic novelty but in a socio-technical integration blueprint for SMEs. Specifically, the innovation involves: (i) integrating a data pipeline (ERP/CRM data lake) with AutoML platforms and interactive dashboards within a coherent data governance workflow; (ii) a human–AI decision support design that enables managers without technical expertise to select appropriate quantile scenarios (P50/P80) and understand forecast implications through a conversational interface; and (iii) an empirical demonstration of feasibility and practical constraints when implementing cloud-based ML in an SME with limited IT resources. The use of established cloud services (AWS) means that the algorithmic components themselves are not novel; rather, the contribution is the accessible and reproducible implementation architecture.

The application's level of innovation is justified by making it easier for SMEs to use ML to develop predictive models that forecast product demand. In demand forecasting, ML can analyze historical data on sales, prices, promotions, and other factors to identify patterns that help predict future demand.

Using ML for demand forecasting offers several advantages over traditional methods. First, it provides greater accuracy by considering a wide range of factors and identifying patterns that are invisible to the human eye. Additionally, machine learning is faster, enabling companies to make decisions quickly and respond promptly to market changes.

An analysis of predictive model functionality indicates that this capability is absent from current enterprise resource planning (ERP) systems, which generate reports solely from historical data. Likewise, business intelligence (BI) systems display only historical data, not data for predictive analysis. Although specialized systems that cross-reference data and offer predictive models exist, their high development costs make this technology inaccessible to small and medium-sized enterprises (SMEs). Implementing such technologies requires experts like data scientists, AI specialists, and cloud computing professionals, highly paid professionals that many SMEs do not have.

Carvalho et al. [7] highlight the challenge of finding these IT professionals in Brazil and the shortage of trainees. The application described in this practical paper aims to address these gaps by offering a simple and user-friendly interface that provides advanced computational resources to SMEs. This feature will enable these companies to leverage advanced AI to solve major business problems, democratizing data analytics and making it accessible to more companies that might not be able to afford high implementation costs. The solution's intuitive and straightforward interface enables managers without extensive technical knowledge to use the tool effectively, conducting simulations and clearly understanding forecasts.

Another key benefit is the system's customization and flexibility, which can be tailored to company-specific data and continuously improved as new data sources become available. Combining data from various internal and external sources through an ETL (Extract, Transform, Load) process offers a comprehensive and precise view for decision-making.

Complexity

Developing a ML product demand prediction project is quite complex, needing close teamwork and diverse expertise. It involves collaboration among data scientists, software engineers, supply chain specialists, and project managers. The involvement of external stakeholders like suppliers and customers is essential for collecting relevant data and verifying the predictive models, offering insights beyond internal sources [25].

The project requires advanced technical expertise in ML, including supervised and unsupervised learning algorithms, data preprocessing techniques, feature engineering, and model validation [11]. Furthermore, using collaboration and project management tools can improve communication and cooperation among participants, helping identify uncertainties, risks, and critical factors for the project's success [4].

When implementing the project in other companies, some risks should be considered, including poor planning, a lack of technical skills, difficulties with system integration, and employee resistance to change. Integrating with existing systems can lead to problems such as incompatibility, failures, and errors. Additionally, employee resistance to adopting the new ML application can lead to mistakes, misuse, and low adoption.

Specific technological risks include data quality, as incomplete, inaccurate, or biased data can lead to inefficient or incorrect predictions. Information overload can make data management and interpretation difficult. Data security is essential to prevent breaches. The performance of an AI model may not reach the desired accuracy due to technical limitations or insufficient data. Rapid technological advances can quickly render the system obsolete, requiring continuous updates and ongoing investments.

Additionally, reliance on appropriate technological infrastructure is necessary to support data processing and storage. A dependence on IT vendors, such as AWS, that can migrate to another platform or even render their own platform obsolete, could increase costs and pose a risk to the project. Regarding regulatory risks, compliance with data protection laws such as the Brazilian LGPD (General Data Protection Law) is critical to avoid fines and reputational damage. There are also specific regulations on AI use and industry standards that must be followed to ensure security and compliance. To address these risks, strategies such as proper planning, training and capacity building, planned integration, stakeholder engagement, data governance, technological updates, and cybersecurity measures should be adopted.

Discussion of results

Theoretical implications

The results provide preliminary evidence that using machine learning techniques, enabled by cloud platforms, can enhance demand forecasting and inventory management at a building materials distributor. The achieved WAPE of approximately 0.69% represents the aggregate error during the evaluation period, though this should be interpreted with caution: (i) the metric is volume-weighted and may be influenced by high-turnover items; (ii) a systematic comparison with classical baselines (naïve, seasonal naïve) was not conducted; and (iii) the error distribution across SKU classes and forecast horizons was not analyzed.

Accordingly, the primary value of this study should not be assessed through the lens of forecasting accuracy alone. The contribution is fundamentally architectural and socio-technical since it demonstrates how an SME that previously relied exclusively on descriptive analytics—historical sales reports generated by its ERP system—can transition to a simplified predictive analytics capability using cloud-based AutoML platforms. For Casa Cardão, this leap is analogous to moving from a basic point-and-shoot camera to a professional camera with guided settings. Analogously, the solution does not represent the most sophisticated approach in absolute terms, but it provides a relevant, accessible, and "palatable" capability gain for the SME context. Critically, the implemented architecture enables the company to conduct future analyses autonomously, without permanent dependence on the research team, thereby contributing to organizational learning and digital maturity. The novelty of the paper is associated

with its demonstration of a feasible path for SMEs toward predictive analysis.

The absence of a systematic comparison with naïve or seasonal naïve baselines constitutes a methodological limitation. This omission is partly attributable to the "black box" nature of the AWS AutoML platform, which does not expose its internal benchmarking against elementary baselines or allow users to inject custom algorithms into the evaluation pipeline. AWS platform employs its own proprietary algorithms (such as DeepAR+, a variation of LSTM networks) and performs automatic hyperparameter optimization. This automation is constitutes a deliberate trade-off between methodological control and implementation practicality. Future research should address this gap by conducting parallel evaluations using open-source frameworks that allow full baseline comparisons under identical evaluation protocols.

This approach is expected to work best when: (i) sufficient historical data (16+months) is available; (ii) demand patterns include seasonal or complex components that traditional methods struggle to capture; and (iii) the organization lacks specialized data science staff but has basic IT capability to manage cloud integrations. The approach may be less suitable for: (i) highly intermittent demand with frequent zero-demand periods; (ii) products with short life cycles or frequent specification changes; (iii) contexts where data quality is poor or historical records are incomplete; and (iv) organizations requiring full algorithmic transparency for regulatory or governance reasons.

The cloud-based AutoML approach involves several trade-offs that organizations should consider: (i) Accuracy vs. Explainability: While AutoML can achieve low error rates, the underlying model (e.g., DeepAR+neural networks) functions as a "black box," limiting the ability to explain individual predictions. The conversational dashboard mitigates this by providing scenario explanations, but full algorithmic transparency is not achieved. (ii) Cost vs. In-house Capability: The estimated monthly TCO of USD 500–1500/month[1] avoids the higher cost of hiring specialized data scientists but creates ongoing operational expenses and vendor dependency. (iii) Convenience vs. Vendor Lock-in: AWS-managed services simplify implementation but create dependency on a single vendor's ecosystem, pricing, and product lifecycle decisions (as illustrated by the Amazon Forecast transition). (iv) Speed vs. Customization: AutoML enables rapid deployment but limits fine-grained control over hyperparameters and model architectures.

This finding aligns with the literature highlighting the benefits of ML for capturing complex patterns in demand time series [15, 22]. The ability of the algorithms used (such as DeepAR+) to automatically incorporate features like seasonality, trends, and related metadata (if provided) was crucial to the performance achieved.

Using platforms such as Amazon Forecast and SageMaker Canvas proved an effective way to overcome barriers to ML adoption, especially for SMEs such as Casa Cardão. Automating tasks such as algorithm selection, hyperparameter tuning, and model evaluation (AutoML) accelerated development and enabled strong results without requiring a large team of data scientists [1, 2].

The tangible benefits observed—reduction in stockouts, inventory optimization, and improved planning—corroborate the positive impact of technology on operational efficiency and potential cost savings [6, 8]. By minimizing waste (associated with excess inventory) and lost sales (due to disruptions), the solution directly contributed to the company's competitiveness.

Managerial recommendations

From a decision-making perspective, companies must balance the trade-offs among processing costs, predictive performance, and explainability when selecting cloud platforms. Services like Amazon SageMaker Canvas offer competitive pay-as-you-go pricing and require minimal setup for data pipelines. In contrast, open-source solutions (e.g., Prophet) offer greater algorithmic transparency but depend on an internal data science team. For SMEs with limited IT resources, Casa Cardão's experience indicates that outsourcing infrastructure is an efficient and replicable way to gradually adopt AI in operations.

Based on our implementation experience, the estimated monthly cost of maintaining the AWS infrastructure (S3 storage, Lambda functions, SageMaker Canvas training and inference) ranges from USD 500 to USD 1,500, depending on data volume, the number of SKUs analyzed, and retraining frequency. This budget compares favorably with hiring dedicated data science personnel (annual salaries in Brazil for qualified professionals exceed USD 60,000), though organizations should factor in ongoing cloud costs when assessing long-term sustainability beyond initial grant-funded or credit-supported phases.

[1] *Monthly operational costs (post-implementation):*—Amazon S3 storage: $50–100 (depending on data volume)—AWS Lambda functions: $20–50 (for automation and data processing)—SageMaker Canvas: $300–1,000 (based on training frequency and data volume)—Data Wrangler: $50–150 (for preprocessing)—CloudWatch monitoring: $20–50—**Estimated total: $500–1,500/month.**

The implemented solution includes several governance mechanisms: (i) data governance: All data are stored in versioned S3 buckets with access logging; (ii) access control: role-based access within AWS IAM ensures that only authorized personnel can modify models or access sensitive data; (iii) model monitoring: The 5% WAPE threshold triggers automatic retraining, with monitoring logs retained for audit purposes; and (iv) mitigation of over-reliance: The system presents multiple quantile scenarios (P50, P60, ande P80) rather than single-point forecasts, encouraging managers to exercise judgment rather than blindly accepting algorithmic recommendations. Nevertheless, a comprehensive organizational change assessment (user adoption, learning curves, and impact on decision-making culture) was not conducted and represents an avenue for future research.

However, some limitations and challenges arose. The quality and availability of historical data were essential; gaps or inconsistencies required significant effort during preprocessing. Additionally, the interpretability of more complex ML models, known as "black box" models, can be difficult, though platforms like SageMaker Canvas offer tools to help explain their predictions. Successful implementation also depended on a cultural shift and on adapting internal company processes to build trust and effectively use the predictions generated by the system. Despite these challenges, the experience demonstrates that integrating AI into SCM processes is practical and beneficial.

Conclusion

This practice and policy paper describes and analyzes the development and deployment of a cloud-based AutoML demand forecasting system at Casa Cardão, a medium-sized Brazilian distributor of building materials. The study's primary contribution is architectural and socio-technical rather than methodological or algorithmic. The originality of the studies lies in providing a detailed and reproducible implementation blueprint, showing how an SME that previously relied solely on descriptive analytics (historical ERP and BI reports) can transition to a simplified predictive analytics capability using managed cloud platforms (Amazon Forecast and SageMaker Canvas). The implemented system achieved a WAPE of approximately 0.69% during the evaluation period, though this aggregate metric should be interpreted cautiously given the limitations discussed below. Preliminary qualitative observations suggest potential benefits in reducing stockouts, optimizing inventory levels, and improving purchasing planning, though systematic pre- and post-measurement of quantitative business KPIs was not conducted and requires longer evaluation periods. The study illustrates how cloud-based AutoML platforms can lower barriers to entry for SMEs that lack specialized Data Science teams, providing a concrete and exploratory example of how AI-driven demand forecasting can support digital transformation in supply chain management. This case study may be a motivator for similar SMEs willing to leverage their analytic capabilities.

From both theoretical and practical perspectives, the study contributes by detailing an end-to-end ML implementation in an SME supply chain, including the technology architecture, data pipeline, and human–machine interaction protocols. It reinforces evidence that advanced ML models, when combined with managerial judgment and appropriate performance monitoring, can capture nonlinear and seasonal patterns that traditional statistical methods may miss, particularly in volatile demand environments such as construction supply distribution. The project also highlights the importance of governance mechanisms—such as continuous accuracy monitoring, retraining triggers, and feedback loops between forecasts and operational decisions—to sustain predictive performance over time.

This study has several limitations that should be acknowledged. First, as a single-case practice study, the results are context-dependent and should not be generalized without validation in other settings. Second, only aggregate WAPE is reported; per-SKU class breakdowns, intermittent demand analysis, and multi-horizon error distributions were not conducted. Third, a systematic comparison with naïve and seasonal naïve baselines under identical evaluation protocols was not performed. Fourth, business impact claims (inventory reduction and stockout reduction) are presented as expected benefits based on preliminary qualitative observations rather than as empirically confirmed achievements. Fifth, reliance on AWS AutoML limits methodological transparency, as specific hyperparameters were neither manually configured nor reported. Sixth, the cold-start problem for new products without historical data remains unaddressed. Seventh, user adoption and organizational change dimensions were not systematically assessed. Eighth, data cannot be made publicly available due to confidentiality agreements, limiting reproducibility. Finally, the solution benefited from temporary AWS credits, and long-term financial sustainability at the estimated TCO of USD 500–1500/month requires validation.

Several of these limitations merit further contextualization. The absence of baseline comparisons (third limitation) stems primarily from the "black box" nature of the AWS AutoML platform, which does not allow users to inject custom algorithms into its evaluation

pipeline or expose how the selected model compares against elementary baselines such as naïve persistence. This opacity is an inherent feature of proprietary AutoML services and represents a trade-off that the study consciously accepted in exchange for the feasibility of rapid implementation in a resource-constrained SME. Similarly, the limited hyperparameter transparency (fifth limitation) reflects the design philosophy of AutoML platforms, which automate hyperparameter optimization as their core value proposition. The evaluation scope (second limitation) was constrained to 10 representative SKUs from the company's portfolio of over 10,500 items; while these SKUs were selected to represent different ABC classification categories, a comprehensive evaluation across all demand profiles was beyond the scope of this exploratory study. The absence of quantitative business impact measurement (fourth limitation) reflects the study's focus on technical feasibility and implementation architecture rather than on a full cost–benefit analysis, which would require multiple years of post-deployment data collection. While significant, these limitations do not undermine the study's core contribution, which is the description and analysis of an accessible and reproducible implementation blueprint for SME contexts.

Further limitations stem from the technology choices. The solution relied primarily on AWS-managed services (Amazon Forecast and SageMaker Canvas) and benefited from temporary cloud credits, which may not reflect the long-term cost structure faced by all SMEs. There is also vendor lock-in and opacity associated with proprietary algorithms, which can limit transparency, reproducibility, and portability to other environments. Additionally, the evaluation period for business outcomes was relatively short and focused mainly on forecasting metrics and qualitative managerial perceptions, rather than on a full cost–benefit analysis that included financial indicators such as return on investment, working capital reduction, or service-level improvements over multiple years. Finally, the study did not systematically assess organizational change dimensions, such as user adoption, learning curves, and the impact of the system on roles, routines, and decision-making culture.

These limitations open several opportunities for future studies. One important direction is to design, implement, and evaluate an open-source reference model that replicates the solution's core functionalities at even lower marginal costs and greater transparency. Such a model could be built with open-source libraries and frameworks and released under a permissive license, enabling replication, adaptation, and peer review by researchers, practitioners, and policymakers. Comparative studies could then benchmark proprietary cloud services against fully open-source stacks—considering not only predictive accuracy but also total cost of ownership, explainability, maintainability, and ease of integration with existing ERP and CRM systems.

Future research should also extend the empirical scope beyond a single company and sector. Multi-case or cross-industry studies—including food retail, pharmaceuticals, automotive parts, and other sectors with complex, intermittent demand—could test the proposed architecture's robustness and identify contingencies that moderate its effectiveness (e.g., product variety, demand volatility, lead times, or data availability). Longitudinal designs could assess the long-term financial impact of ML-based demand forecasting on inventory turnover, stockout rates, service levels, and profitability, ideally using quasi-experimental designs or A/B tests comparing business units with and without the predictive system.

Another promising line of work integrates demand forecasting with optimization models and prescriptive analytics. Rather than stopping at forecasting, future applications could combine predictions with optimization routines for order quantities, safety stocks, and replenishment policies, forming closed-loop decision support systems. Research could also delve more deeply into socio-technical aspects, including user trust in algorithmic recommendations, the design of explainable interfaces, training programs for managers, and governance frameworks for AI in SMEs. Finally, policy-oriented studies could examine how public agencies, development banks, and innovation programs might support the diffusion of low-cost, open, and responsible AI solutions for inventory management among small and medium-sized firms.

This study concludes that advanced cloud-based ML techniques can substantially improve demand forecasting and inventory management for a medium-sized construction supplies distributor, but it also warns that such solutions are not off-the-shelf panaceas. Their success depends on data availability and quality, careful technology choices, sustained organizational engagement, and appropriate governance. Recognizing these limitations and exploring the proposed possibilities for future work may help researchers, practitioners, and policymakers design more robust, transparent, and accessible predictive systems—contributing to the broader goal of democratizing AI-enabled supply chain optimization for SMEs.

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Author contributions
The research group's coordination and the primary manuscript's composition were undertaken by D.D.G. R.B.C. served as the doctoral supervisor and

performed a comprehensive review of the document. M.A.H., in conjunction with B.K.G.N.C., bore responsibility for the application's development and ongoing support during the Proof of Concept phase, in addition to authoring the front-end interface.

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Data availability
The transactional data used in this study cannot be made publicly available due to confidentiality agreements with the partner company. The methodology, technological architecture, and implementation procedures are fully described in this manuscript to enable methodological replication. Researchers seeking to validate or extend this work may develop synthetic datasets based on the described characteristics or apply the documented architecture to their own organizational data.

Declarations

Ethics approval and consent to participate
This study did not involve human participants, personal data, or interventions requiring ethical approval. The research used organizational and operational data provided by a private company under a confidentiality agreement. Therefore, ethical approval and consent to participate are not applicable.

Consent for publication
All authors have reviewed and approved the final version of the manuscript and consent to its publication.

Competing interests
The authors declare no competing interests.

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